THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 2310	Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$
2	**(c)** 28	$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
3	**(d)** -4	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
4	**(b)** 29	$(\alpha+\beta)^2 - 2\alpha\beta$
5	**(a)** 0	Common difference is -1; $a_{p+q} = 0$
6	**(c)** 8	$(m+n)^2 - m^2$
7	**(c)** 12	Let the point be $(0, y)$
8	**(d)** $\frac{25}{9}$	$1 + \tan^2\theta$
9	**(c)** $2+2\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 4$ gives $h = 4 \div (\sqrt{3}-1)$; Height $= 2+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m)
10	**(d)** 48 cm	$AM = \frac{30\times40}{50} = 24$ cm; $AB = 2AM = 48$ cm
11	**(c)** 4.47 cm²	Sector $= 25.67$ cm²; triangle $= \frac{\sqrt{3}}{4}r^2 = 21.19$ cm²; Segment $= 25.67 - 21.19 = 4.47$ cm²
12	**(d)** 4410 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851$ cm³; cube $= 9261$ cm³; Wasted $= 4410$ cm³
13	**(b)** 23	$\frac{4325+25p}{77+p} = 49$ gives $(25-49)p = -552$; $p = 23$
14	**(d)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
15	**(d)** 60	LCM of 1 to 6
16	**(b)** 30	$\alpha\beta(\alpha+\beta)$
17	**(d)** 2	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
18	**(c)** 24	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
19	**(c)** 54	$a_n = \frac{2S_n}{n} - a$
20	**(a)** 6	Find b by Pythagoras
21	**(c)** 63	Twice the area
22	**(b)** $-\frac{7}{25}$	$\cos^2\theta - \sin^2\theta$
23	**(c)** 20 m	Height of the top of the flagstaff $= d\tan 60^\circ = 30$ m; Length $= 30 - 10 = 20$ m
24	**(b)** 14 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 7 = 14$ cm
25	**(b)** 150 mm	Circumference $= \pi d = 66$ mm; diameters $= 4 \times 21 = 84$ mm; Total $= 150$ mm
26	**(c)** 2425.5 cm³	Cylinder $= \pi r^2(2r) = 7276.5$ cm³; sphere $= \frac{4}{3}\pi r^3 = 4851$ cm³; Unoccupied $= 2425.5$ cm³
27	**(b)** 9	$238.4 = 230 + \frac{50-(20+x)}{25} \times 10$ gives $x = 9$; $x = 9$; then $y = 100 - 82 - x = 9$
28	**(b)** $\frac{1}{6}$	Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
29	**(b)** 9	HCF of the differences from the remainder
30	**(b)** 7	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
31	**(b)** 8	Solve for x and y
32	**(a)** 6	$D = 0$ gives $k^2 = 4m$
33	**(a)** 46	$a_n = S_n - S_{n-1}$
34	**(b)** $\frac{12}{5}$	$\frac{1}{2}ab = \frac{1}{2}ch$
35	**(c)** $\frac{16}{3}$	Equate squared distances
36	**(a)** $\frac{24}{25}$	$2 \cdot \frac{p}{c} \cdot \frac{q}{c}$
37	**(c)** 2 minutes	Distance covered in 4 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 4 \div 2 = 2$ minutes

38	**(a)** 49 cm	$15x + 4 = 8x + 25$ gives $x = 3$; $QA = 15(3) + 4 = 49$ cm
39	**(c)** 1925 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 35^2$; Area $= 1925$ m^2
40	**(b)** 15 cm	$H = 3h = 3 \times 5$; $H = 15$ cm
41	**(d)** 269.8	$\sum f_i = 100$ and $\sum f_i x_i = 26980$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{26980}{100} = 269.8$
42	**(a)** 45	$\frac{x}{15+x} = 3 \times \frac{15}{15+x}$ gives $x = 3 \times 15$; $x = 45$
43	**(d)** 3	$12 = 2^2 \times 3$; multiply by the primes that occur to an odd power
44	**(a)** $\frac{5}{2}$	$\dfrac{\alpha^2 + \beta^2}{\alpha\beta}$
45	**(b)** 12	Intercepts 6 and 4; area $= \frac{1}{2} \times 6 \times 4$
46	**(a)** 2	Substitute the root
47	**(d)** 4	$d = 2p$
48	**(d)** 14	Sides in ratio $a : b$
49	**(d)** 6	$x = 0$: $mc + na = 0$
50	**(b)** $\frac{1}{3}$	$(\sec + \tan)(\sec - \tan) = 1$

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** 12 m	Both objects are on the same side, so the distances are subtracted; Distance $= 12$ m
2	**(b)** 120 cm	Half-chord $= \sqrt{68^2 - 32^2} = 60$ cm; Chord $= 2 \times 60 = 120$ cm
3	**(b)** 117 cm	Arc $= \frac{45}{360} \times 2\pi \times 42 = 33$ cm; Perimeter $= 33 + 84 = 117$ cm
4	**(b)** 14 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 14$ cm
5	**(b)** 108	Median class 100–120: $l = 100$, $cf = 44$, $f = 15$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 100 + \frac{50-44}{15} \times 20 = 108$
6	**(c)** 5	Cross-multiplying: $7(7 + x) = 4(16 + x)$; $x = 5$
7	**(c)** $\frac{\sqrt{12}}{\sqrt{3}}$	$\sqrt{12}/\sqrt{3} = 2$
8	**(b)** 11	$\frac{c}{a}$ for the monic cubic
9	**(a)** 29	$(x + y)^2 - 2xy = 49 - 20$
10	**(c)** $k \leq 3$ and $k \neq 0$	$D = 36 - 12k \geq 0$
11	**(a)** 1325	Find n first
12	**(c)** $\frac{9}{5}$	$AB^2 = AD \cdot AC$
13	**(d)** 50	$s^2 = d^2/2$
14	**(d)** 1	Each pair sums to 1
15	**(a)** 4 m	Part of the tower above the top of the building $= d \tan 30^\circ = 1$ m; Tower $= 4$ m
16	**(b)** 15 cm	$AF = AB - r = 45 - r$, $CD = BC - r = 60 - r$ and $AF + CD = AC$ (tangent lengths); $(45 - r) + (60 - r) = 75$ gives $r = 15$ cm
17	**(c)** 30°	$\theta = \frac{22 \times 360}{2 \times \frac{22}{7} \times 42}$; $\theta = 30^\circ$
18	**(c)** 4 : 9	$2^2 : 3^2 = 4 : 9$; Ratio $= 4 : 9$
19	**(c)** 63	$l = 60$, $f_1 = 24$, $f_0 = 21$, $f_2 = 17$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{3}{10} \times 10 = 63$
20	**(c)** $\frac{4}{5}$	Favourable outcomes $=$ number of red balls; $P = \frac{20}{25} = \frac{4}{5}$
21	**(a)** $2^3 \times 5^2 \times 3$	Zeros at the end = min(power of 2, power of 5)
22	**(b)** 21	$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$
23	**(c)** 3	Subtract the equations
24	**(b)** $\frac{5}{6}$	$\frac{\alpha + \beta}{\alpha\beta}$
25	**(b)** 11	$S_n = 0 \Rightarrow n = 1 - \frac{2a}{d}$
26	**(a)** 25	Side $= s$
27	**(d)** 28	Use the area formula
28	**(a)** 1	Pair as $\tan\theta \tan(90^\circ - \theta) = 1$
29	**(d)** $9 + 3\sqrt{3}$ m	The points are on opposite sides, so the distances add; Distance $= 9 + 3\sqrt{3}$ m
30	**(a)** 36 cm	$PT^2 = OP^2 - OT^2 = 85^2 - 77^2 = 1296$; $PT = 36$ cm
31	**(d)** 1400 cm²	Circle $= 3850$ cm^2; square $= 2450$ cm^2; Difference $= 1400$ cm^2
32	**(a)** 8 : 27	$2^3 : 3^3 = 8 : 27$; Ratio $= 8 : 27$
33	**(c)** 25	$20 + \frac{25-18}{14} \times 10 = 25$
34	**(d)** $\frac{19}{70}$	Favourable numbers: 19 outcomes; number $= 19$; $P = \frac{19}{70}$
35	**(a)** 144	Highest powers of $2^2 \times 3^2$, $2^4 \times 3$, $2^3 \times 3^2$

36	**(b)** 7	Sum of zeroes $= 9$, so other zero $= 9 - 2$
37	**(b)** -6	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
38	**(a)** 53	$(\alpha + \beta)^2 - 2\alpha\beta$
39	**(b)** 0	Common difference is -1; $a_{p+q} = 0$
40	**(b)** 21	$(m + n)^2 - m^2$
41	**(b)** 9	Let the point be $(0, y)$
42	**(b)** $\frac{169}{25}$	$1 + \tan^2\theta$
43	**(b)** 30°	$\tan\theta = \frac{\sqrt{3}}{3}$ which is $\tan 30^\circ$; $\theta = 30^\circ$
44	**(c)** the perpendicular bisector of AB	$PA = PB$ and $OA = OB$
45	**(c)** 924 m²	Increase $= \frac{1}{4} \times \frac{22}{7}(35^2 - 7^2)$; Increase $= 924$ m^2
46	**(b)** 30	$\pi(6)^2(10) = n \times \frac{1}{3}\pi(3)^2(4)$; $n = 30$
47	**(a)** 20	$\frac{14660 + 190p}{80 + p} = 184.6$ gives $(190 - 184.6)p = 108$; $p = 20$
48	**(c)** $\frac{43}{50}$	Favourable outcomes = number of good pens; $P = \frac{215}{250} = \frac{43}{50}$
49	**(b)** 420	LCM of 1 to 7
50	**(b)** 72	$(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 14	Needs $\frac{k}{4p} = \frac{p}{k} = \frac{k-p}{k}$
2	**(c)** 32	$(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$
3	**(d)** 59	$a_n = \frac{2S_n}{n} - a$
4	**(b)** 30	Find b by Pythagoras
5	**(b)** 44	Twice the area
6	**(b)** $-\frac{119}{169}$	$\cos^2\theta - \sin^2\theta$
7	**(c)** $3+3\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 6$ gives $h = 6 \div (\sqrt{3}-1)$; Height $= 3+3\sqrt{3}$ m (distances $9+3\sqrt{3}$ m and $3+3\sqrt{3}$ m)
8	**(d)** 26.88 cm	$AM = \frac{14\times 48}{50} = 13.44$ cm; $AB = 2AM = 26.88$ cm
9	**(b)** 67.82 cm²	Sector $= 115.50$ cm²; triangle with sides r, r and angle 120° has area $\frac{1}{2}r^2\sin 120^\circ = \frac{\sqrt{3}}{4}r^2 = 47.68$ cm²; Segment $= 115.50 - 47.68 = 67.82$ cm²
10	**(b)** 643.5 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3}\times\frac{22}{7}(9^3 - 7.5^3)$; Volume $= 643.5$ cm³
11	**(c)** 21	$25 = 20 + \frac{50-(20+x)}{18}\times 10$ gives $x = 21$; $x = 21$; then $y = 100 - 75 - x = 4$
12	**(c)** $\frac{1}{6}$	Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
13	**(d)** 24	HCF of the differences from the remainder
14	**(c)** 56	$\alpha\beta(\alpha+\beta)$
15	**(c)** 7	Solve for x and y
16	**(d)** 8	$D = 0$ gives $k^2 = 4m$
17	**(c)** 54	$a_n = S_n - S_{n-1}$
18	**(d)** $\frac{60}{13}$	$\frac{1}{2}ab = \frac{1}{2}ch$
19	**(c)** 4	Equate squared distances
20	**(a)** $\frac{120}{169}$	$2\cdot\frac{p}{c}\cdot\frac{q}{c}$
21	**(b)** 24 m	Height of the top of the flagstaff $= d\tan 60^\circ = 36$ m; Length $= 36 - 12 = 24$ m
22	**(c)** 22 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2\times 11 = 22$ cm
23	**(d)** 497 mm	Circumference $= \pi d = 154$ mm; diameters $= 7\times 49 = 343$ mm; Total $= 497$ mm
24	**(c)** 35280 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{42}{2}\right)^3 = 38808$ cm³; cube $= 74088$ cm³; Wasted $= 35280$ cm³
25	**(b)** 71	$\sum f_i = 100$ and $\sum f_i x_i = 7100$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7100}{100} = 71$
26	**(b)** $\frac{2}{9}$	Favourable outcomes: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4); number $= 8$; $P = \frac{8}{36} = \frac{2}{9}$
27	**(b)** 2	$18 = 2\times 3^2$; multiply by the primes that occur to an odd power
28	**(d)** 7	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
29	**(b)** 6	Intercepts 4 and 3; area $= \frac{1}{2}\times 4\times 3$
30	**(b)** 2	Substitute the root
31	**(b)** 6	$d = 2p$
32	**(d)** 18	Sides in ratio $a : b$
33	**(a)** 8	$x = 0$: $mc + na = 0$
34	**(c)** $\frac{1}{5}$	$(\sec+\tan)(\sec-\tan) = 1$
35	**(d)** 4 minutes	Distance covered in 8 min $= h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}}$; distance left $= \frac{h}{\sqrt{3}}$, which is half of it; Time left $= 8 \div 2 = 4$ minutes
36	**(d)** 43 cm	$12x + 7 = 5x + 28$ gives $x = 3$; $QA = 12(3) + 7 = 43$ cm

37	**(d)** 2772 m²	Area $= \frac{180}{360} \times \frac{22}{7} \times 42^2$; Area $= 2772$ m^2
38	**(b)** 19404 cm³	Cylinder $= \pi r^2(2r) = 58212$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 38808$ cm^3; Unoccupied $= 19404$ cm^3
39	**(c)** 190	Median class 180–200: $l = 180$, $cf = 39$, $f = 22$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 180 + \frac{50-39}{22} \times 20 = 190$
40	**(c)** 44	$\frac{x}{11+x} = 4 \times \frac{11}{11+x}$ gives $x = 4 \times 11$; $x = 44$
41	**(a)** $\frac{\sqrt{75}}{\sqrt{3}}$	$\sqrt{75}/\sqrt{3} = 5$
42	**(a)** $\frac{37}{6}$	$\frac{\alpha^2 + \beta^2}{\alpha\beta}$
43	**(c)** 41	$(x+y)^2 - 2xy = 81 - 40$
44	**(c)** $k \leq 2$ and $k \neq 0$	$D = 16 - 8k \geq 0$
45	**(d)** 1116	Find n first
46	**(a)** $\frac{25}{13}$	$AB^2 = AD \cdot AC$
47	**(b)** 50	$s^2 = d^2/2$
48	**(c)** 1	Each pair sums to 1
49	**(b)** 18 m	Both objects are on the same side, so the distances are subtracted; Distance $= 18$ m
50	**(c)** 112 cm	Half-chord $= \sqrt{70^2 - 42^2} = 56$ cm; Chord $= 2 \times 56 = 112$ cm

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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